Conditional probability

1

a 0.23 **b** 0.73

2

a $x = 0$ **b** $x = 0.2, 0.3$

3

a

	2	4	5
3	5	7	8
7	9	11	12

b $\frac{2}{5}$ **c** 1

4

a

	1	2	4	6
1	1	2	4	6
2	2	4	8	12
3	3	6	12	18
4	4	8	16	24
5	5	10	20	30
6	6	12	24	36

b $\frac{1}{2}$ **c** $\frac{3}{10}$ **d** $\frac{1}{2}$

Dependent and independent events

5

a Independent **b** Dependent **c** Dependent **d** Dependent

6

a Yes **b** No

7 0.2

8

a 0.7 **b** Yes **c** 0.5

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9 $\frac{8}{33}$

10 **a** $\frac{32}{663}$ **b** $\frac{4}{39}$

11 **a** **i** $\frac{33}{16\,660}$ **ii** $\frac{1}{649\,740}$ **b** $2\,598\,960$

12 **a** No **b** $\frac{1}{6}$

13 **a** $\frac{1}{45}$ **b** $\frac{1}{44}$ **c** No **d** Dependent

14 **a** $\frac{1}{51}$ **b** $\frac{1}{50}$ **c** Dependent

15 **a** $\frac{5}{18}$ **b** $\frac{1}{3}$

16 **a** $\frac{1}{64}$ **b** $\frac{11}{850}$

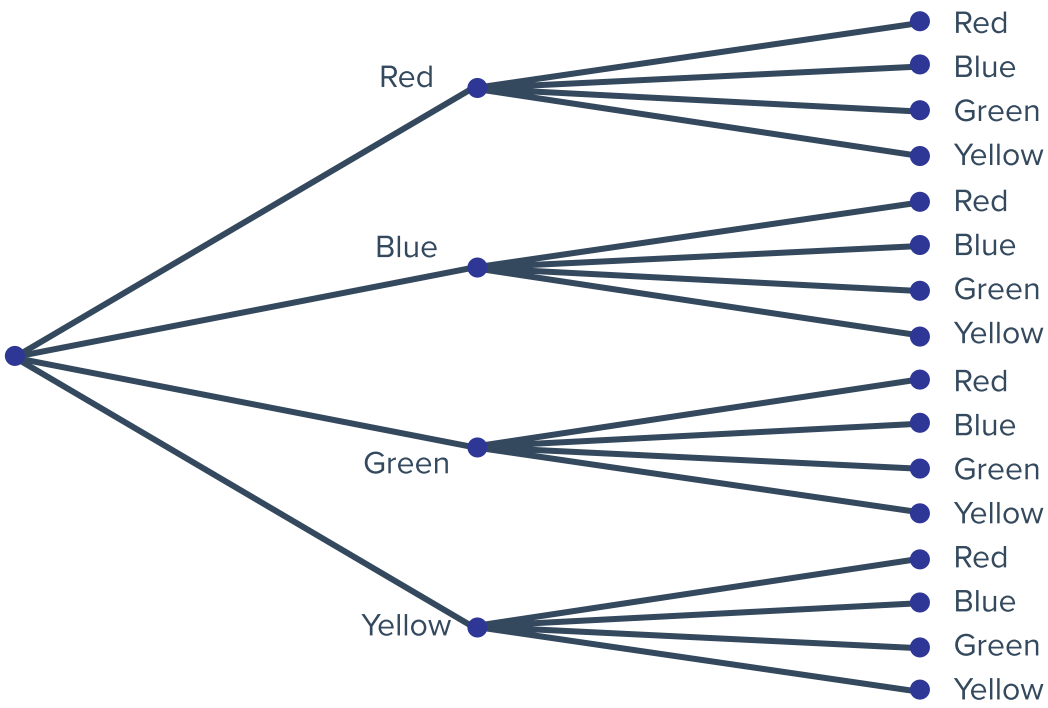
17 **a** $\frac{1}{3}$ **b** $\frac{1}{12}$

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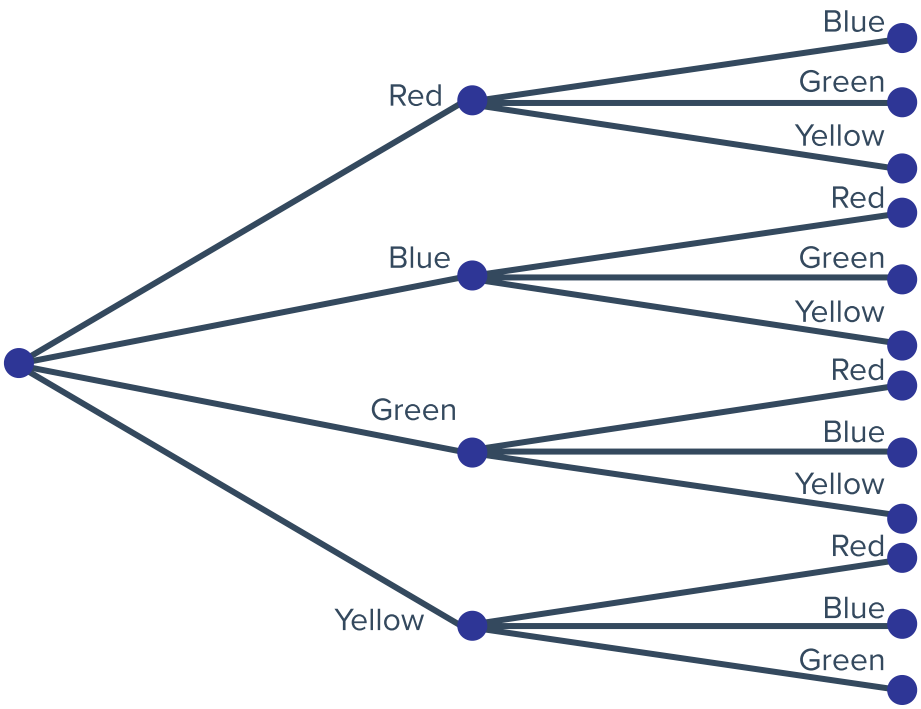
Tree diagrams and probability trees

18

a



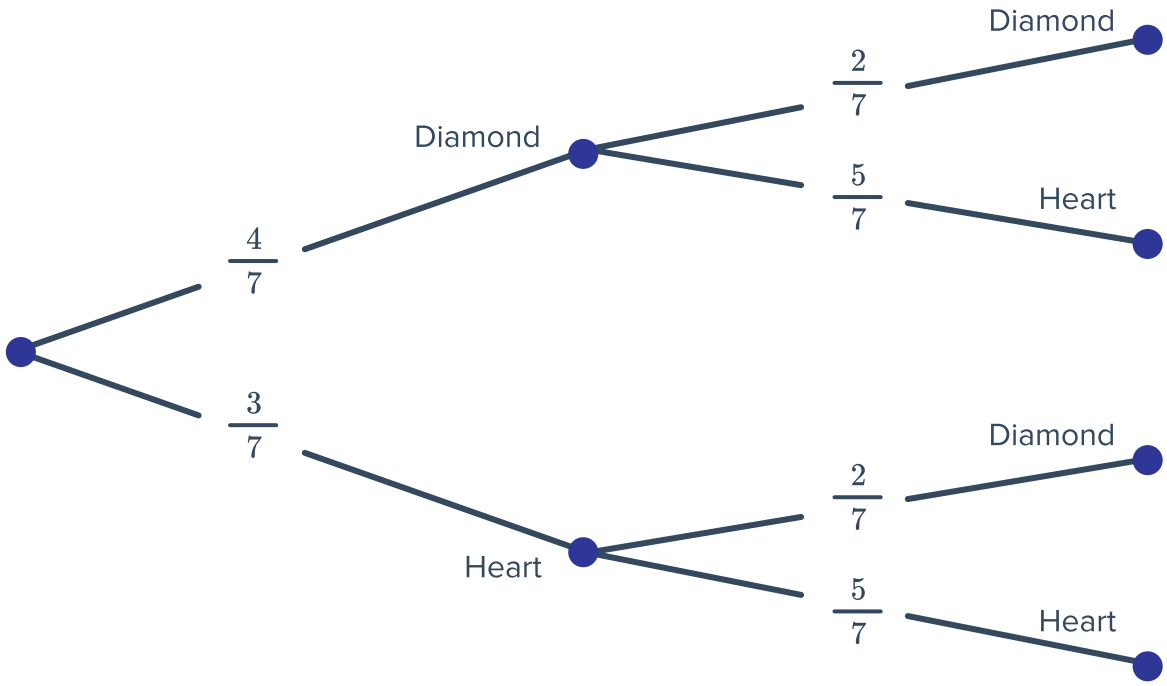
b



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19

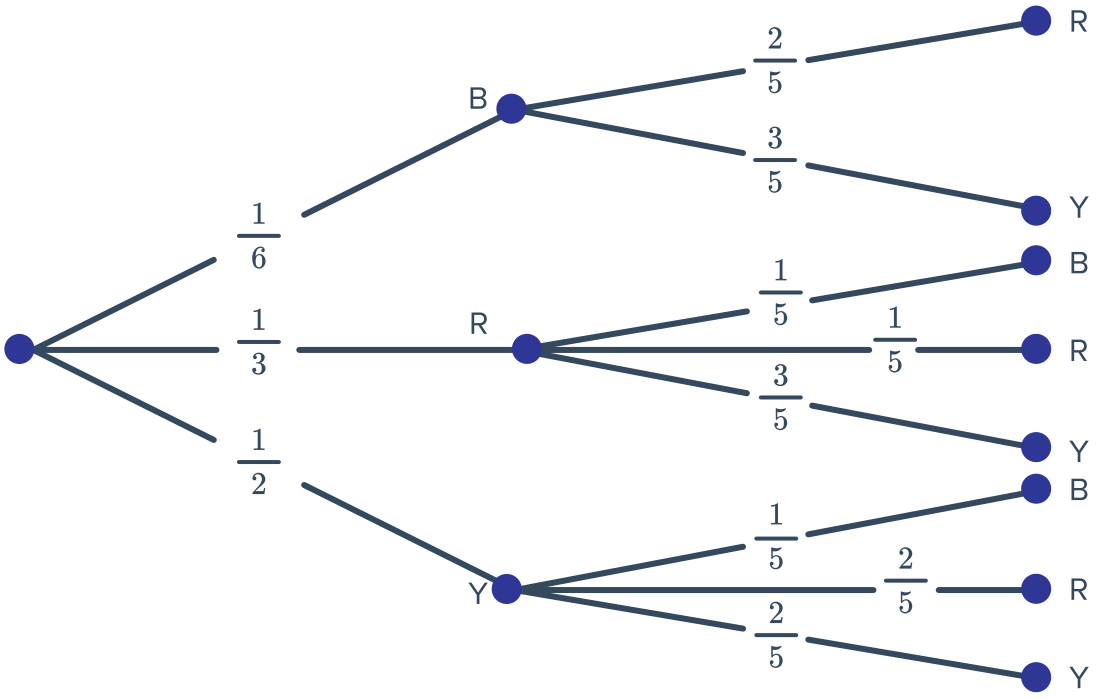
a



b $\frac{15}{49}$

20

a



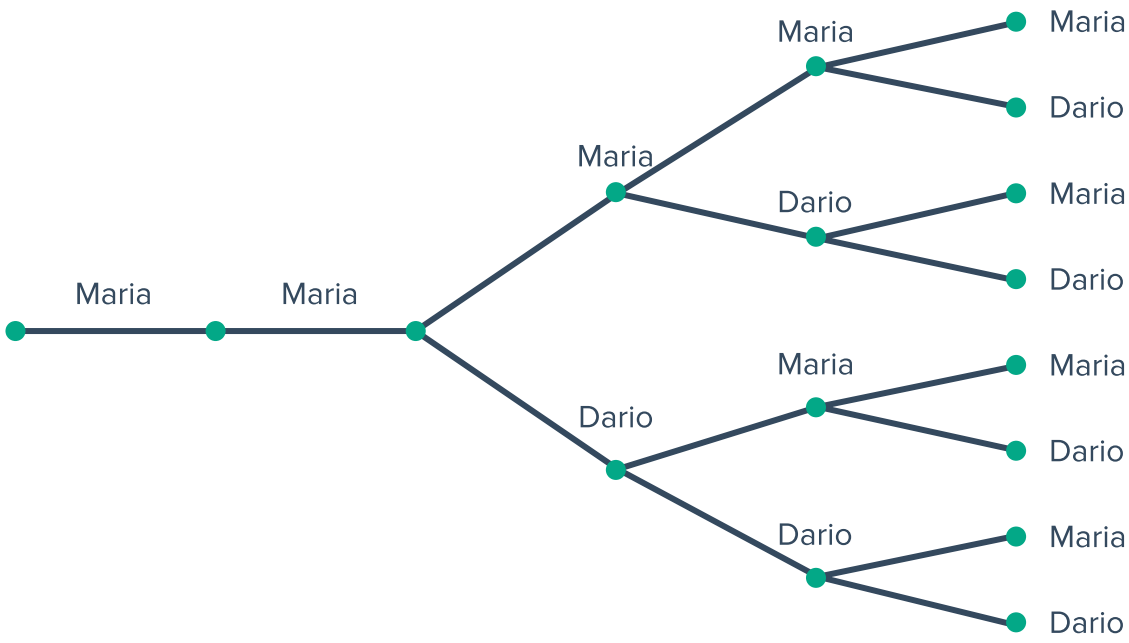
b

i $\frac{1}{10}$ **ii** $\frac{1}{15}$ **iii** $\frac{1}{15}$ **iv** $\frac{1}{5}$

v 0 **vi** $\frac{1}{5}$ **vii** $\frac{2}{5}$

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21 **a**

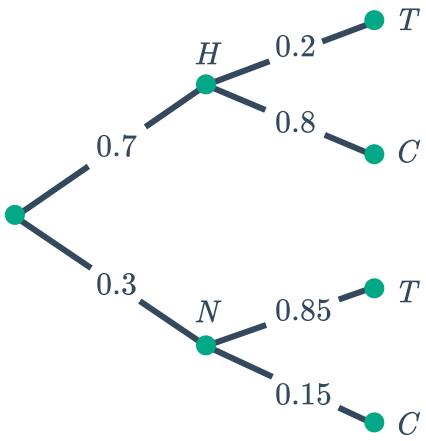


b $\frac{2}{4}$

c $\frac{1}{4}$

22

a

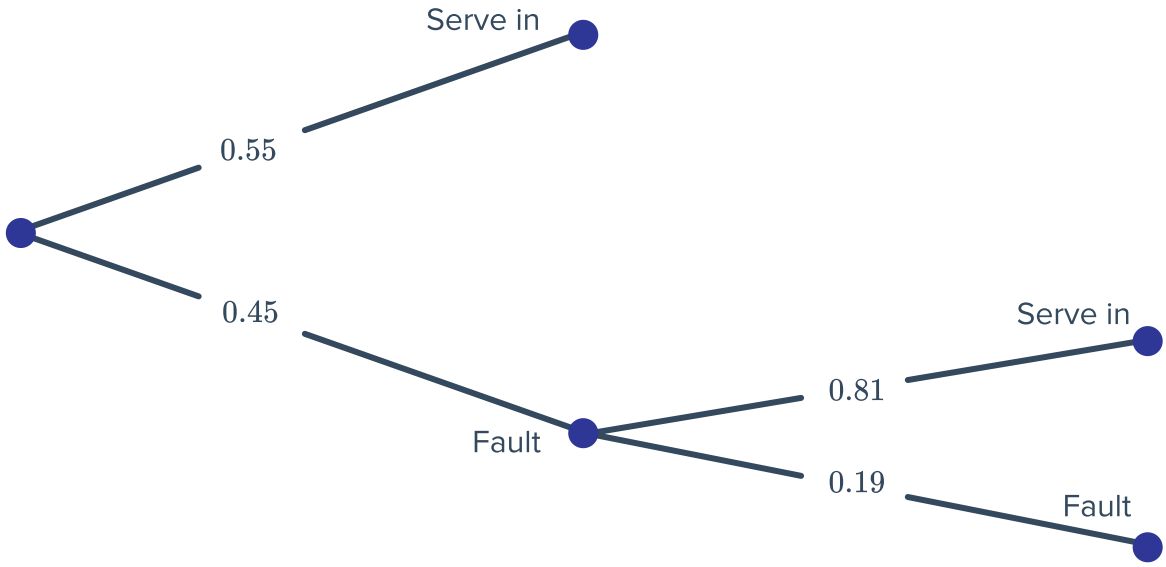


b 0.35

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23

a



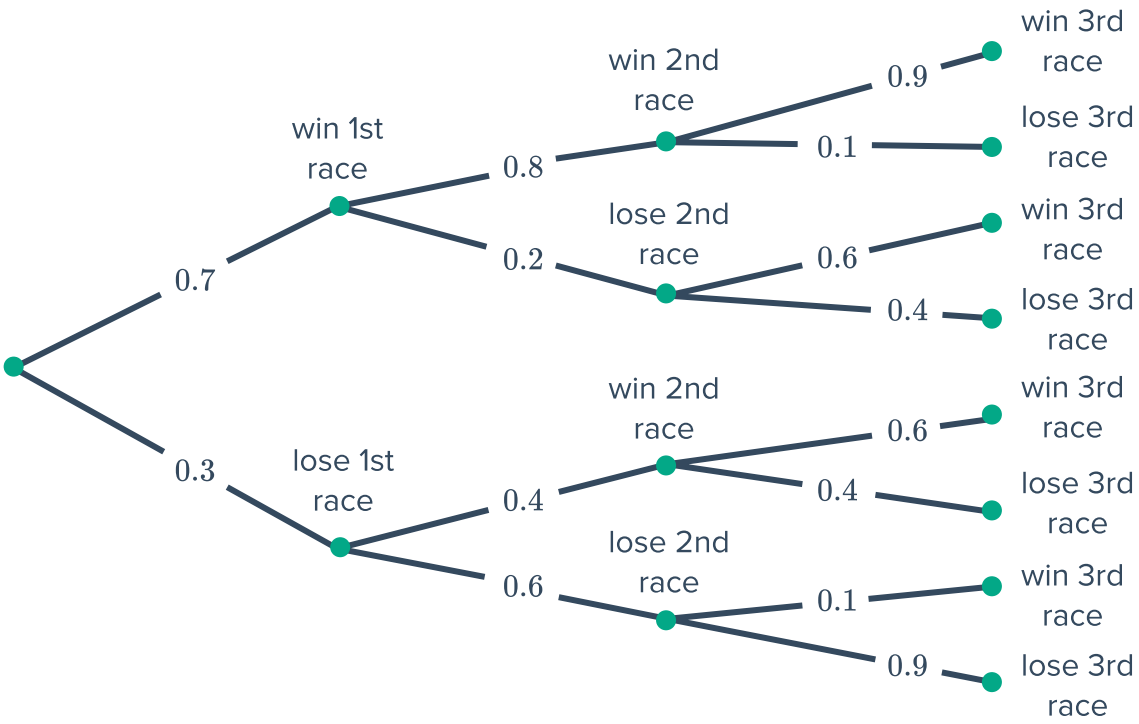
b 0.45

c 0.0855

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26

a



b 0.504

c 0.678

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